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# 6. Case Study: Food Delivery Application – Positive Testing

**Aim:** To identify and analyze the requirements of a Food Delivery Application and prepare positive test scenarios and test cases to verify its functionality.

| Test Case ID | Test Scenario | Values Entered / Action Done | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC01 | User registration | Enter valid user details | Account created successfully | Account created successfully | Pass |
| TC02 | User login | Enter valid credentials | Login successful | Dashboard displayed | Pass |
| TC03 | Search restaurant | Search for “Pizza” | Matching restaurants displayed | Restaurants displayed | Pass |
| TC04 | Select food item | Select an available food item | Food details displayed | Food details displayed | Pass |
| TC05 | Add item to cart | Add a food item | Item added to cart | Item added successfully | Pass |
| TC06 | Update quantity | Increase quantity to 2 | Total price updated | Total price updated | Pass |
| TC07 | Apply coupon | Enter a valid coupon code | Discount applied | Discount not applied | Fail |
| TC08 | Place order | Confirm cart and delivery address | Order placed successfully | Order confirmation displayed | Pass |
| TC09 | Online payment | Use valid payment details | Payment completed successfully | Payment failed unexpectedly | Fail |
| TC10 | Track order | Open order tracking page | Current order status displayed | Order status displayed | Pass |

**Result:** Positive test cases for the Food Delivery Application were prepared and executed using valid inputs. Most functions worked correctly, while coupon application and online payment revealed defects requiring correction.

# 7. Case Study: Food Delivery Application – Negative Testing

**Aim:** To identify and analyze the requirements of a Food Delivery Application and prepare negative test scenarios to verify its behavior under invalid inputs and unexpected conditions.

| Test Case ID | Test Scenario | Values Entered / Action Done | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC01 | Invalid login | Enter an incorrect password | Error message displayed | Error message displayed | Pass |
| TC02 | Empty login fields | Submit without credentials | Validation message displayed | Validation message displayed | Pass |
| TC03 | Invalid restaurant search | Search for an unavailable restaurant | “No restaurants found” message displayed | No restaurants found | Pass |
| TC04 | Unavailable food item | Add an out-of-stock item | Item rejected | Item added to cart | Fail |
| TC05 | Invalid quantity | Enter zero or negative quantity | Invalid quantity rejected | Validation message displayed | Pass |
| TC06 | Invalid coupon | Apply an expired coupon | Coupon rejected | Expired coupon accepted | Fail |
| TC07 | Missing delivery address | Place an order without an address | Address required message displayed | Validation message displayed | Pass |
| TC08 | Payment failure | Simulate a declined payment | Failure displayed and order handled safely | Failure message displayed | Pass |
| TC09 | Duplicate order | Submit the same order repeatedly | Duplicate order prevented | Duplicate order created | Fail |
| TC10 | Network failure | Disconnect during checkout | Error displayed and transaction handled safely | Error displayed; order status remained unclear | Fail |

**Result:** Negative testing identified defects in stock validation, coupon validation, duplicate order prevention, and network failure handling. These issues must be fixed and retested before deployment.

# 8. Case Study: Banking Application – Positive Testing

**Aim:** To identify and analyze the requirements of a Banking Application and prepare positive test scenarios and test cases to verify its major banking operations.

| Test Case ID | Test Scenario | Values Entered / Action Done | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC01 | Valid login | Enter correct customer ID and password | Login successful | Dashboard displayed | Pass |
| TC02 | Balance enquiry | Select balance enquiry | Correct balance displayed | Balance displayed | Pass |
| TC03 | Fund transfer | Transfer ₹1,000 to a valid beneficiary | Transfer successful | Transfer confirmation displayed | Pass |
| TC04 | Cash withdrawal | Withdraw ₹2,000 with sufficient balance | Withdrawal completed | Transaction completed | Pass |
| TC05 | Cash deposit | Deposit ₹5,000 | Balance updated | Balance updated | Pass |
| TC06 | Transaction history | Open transaction history | Previous transactions displayed accurately | Transactions displayed | Pass |
| TC07 | Add beneficiary | Enter valid beneficiary details | Beneficiary added successfully | Beneficiary not added | Fail |
| TC08 | Bill payment | Pay a valid electricity bill | Payment successful | Receipt displayed | Pass |
| TC09 | Change password | Enter valid current and new passwords | Password updated | Password update failed | Fail |
| TC10 | Logout | Select logout | Session terminated | Login page displayed | Pass |

**Result:** Positive test cases were prepared to verify banking operations using valid inputs. Defects were identified in beneficiary registration and password updating, requiring further investigation and correction.

# 9. Case Study: Banking Application – Negative Testing

**Aim:** To identify and analyze the requirements of a Banking Application and prepare negative test scenarios to verify its handling of invalid transactions, unauthorized access, and exceptional conditions.

| Test Case ID | Test Scenario | Values Entered / Action Done | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC01 | Invalid login | Enter incorrect password | Access denied | Error message displayed | Pass |
| TC02 | Empty credentials | Submit without a password | Validation message displayed | Validation message displayed | Pass |
| TC03 | Repeated incorrect login | Enter incorrect credentials repeatedly | Login restricted according to security policy | Access remained available after repeated failures | Fail |
| TC04 | Insufficient balance | Transfer ₹10,000 with ₹5,000 balance | Transfer rejected without debit | Transfer rejected | Pass |
| TC05 | Invalid transfer amount | Enter ₹-500 | Invalid amount rejected | Validation message displayed | Pass |
| TC06 | Invalid beneficiary | Transfer to an invalid account | Transfer rejected | Transfer rejected | Pass |
| TC07 | Exceed transaction limit | Transfer above the permitted limit | Transaction rejected or additional verification required | Transfer processed without required verification | Fail |
| TC08 | Duplicate transaction | Submit the same transfer twice | Duplicate debit prevented | Duplicate debit occurred | Fail |
| TC09 | Unauthorized access | Attempt to access another customer's account | Access denied | Access denied | Pass |
| TC10 | Network failure | Disconnect during a fund transfer | Transaction handled safely without inconsistent balance | Transfer status remained unclear | Fail |

**Result:** Negative testing revealed security and transaction-handling defects involving repeated login failures, transaction limits, duplicate transfers, and network interruptions. These defects require urgent correction because they may affect account security and financial integrity.

# 10. Comprehensive Test Plan: Library Management System

**Aim:** To develop a comprehensive Test Plan for a Library Management System, covering testing strategies, methodologies, and test cases to evaluate its functionality, security, and reliability.

| Test Case ID | Test Scenario | Values Entered / Action Done | Expected Result | Actual Result | Status |
|---|---|---|---|---|---|
| TC01 | Valid login | Enter valid librarian credentials | Dashboard displayed | Dashboard displayed | Pass |
| TC02 | Invalid login | Enter incorrect password | Access denied | Error message displayed | Pass |
| TC03 | Add a book | Enter valid book details | Book added successfully | Book added successfully | Pass |
| TC04 | Search a book | Enter valid title or ISBN | Matching book displayed | Book displayed | Pass |
| TC05 | Register member | Enter valid member details | Member registered successfully | Member registration failed | Fail |
| TC06 | Issue an available book | Select eligible member and available book | Book issued and availability updated | Book issued successfully | Pass |
| TC07 | Issue unavailable book | Select a book with no copies available | Issue rejected or waitlist offered | Unavailable book was issued | Fail |
| TC08 | Return a book | Return an issued book | Loan closed and inventory updated | Loan closed successfully | Pass |
| TC09 | Calculate overdue fine | Return a book after its due date | Fine calculated according to configured rules | Fine calculated incorrectly | Fail |
| TC10 | Renew a book | Renew an eligible loan | Due date updated | Due date updated | Pass |
| TC11 | Duplicate book record | Enter a duplicate ISBN where duplicates are prohibited | Duplicate rejected | Duplicate book record created | Fail |
| TC12 | Unauthorized deletion | Member attempts to delete a book | Access denied | Access denied | Pass |
| TC13 | Invalid member ID | Issue a book to a nonexistent member | Operation rejected | Operation rejected | Pass |
| TC14 | Empty mandatory fields | Submit a blank book form | Validation messages displayed | Validation messages displayed | Pass |

| | | | | | |
|---|---|---|---|---|---|
| TC15 | Generate overdue report | Request an overdue-book report | Accurate overdue records displayed | Report omitted some overdue books | Fail |

**Result:** The Library Management System test plan was prepared with functional test cases covering authentication, book management, member registration, issue and return operations, fine calculation, and report generation. Testing identified defects in member registration, inventory validation, fine calculation, duplicate book prevention, and overdue reports. These defects should be corrected and the affected test cases rerun before the system is approved for release.